Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 31

Elliptic Curve Cryptography-I

Estimated Reading Time: 15–20 minutes

Introduction

Elliptic Curve Cryptography (ECC) is a public-key cryptographic technique that uses the mathematical properties of elliptic curves to provide security. It is widely used in blockchain technology for digital signatures, identity verification, and secure transaction authorization.

ECC provides strong security with comparatively small key sizes, making it suitable for systems where storage, communication, and computational efficiency are important.

What Is an Elliptic Curve?

An elliptic curve is a mathematical curve defined by a specific equation. In cryptography, elliptic curves are generally defined over finite fields, where calculations are performed using a limited set of numbers.

A commonly used form of an elliptic curve equation is:

$$y^2 = x^3 + ax + b$$

Here, x and y represent coordinates, while a and b are curve parameters. The curve must satisfy certain mathematical conditions to be suitable for cryptographic applications.

Unlike ordinary curves used in geometry, cryptographic elliptic curves operate over finite fields, allowing their points to be represented and manipulated using modular arithmetic.

Basic Concepts of ECC

The following concepts are important for understanding ECC:

- Finite field: A mathematical system containing a finite number of elements in which arithmetic operations are performed according to specific rules.
- Curve point: A point whose coordinates satisfy the elliptic curve equation.

Campus : Prestige Vihar, Scheme No. 74-C, Sector D, Vijay Nagar, Indore -452010 (M.P.) India
Phone : 0731-4013333, Fax : 0731-4013329 E-mail : info@piemr.edu.in, Web. www.piemr.edu.in

- Point addition: An operation that combines two points on an elliptic curve to produce another point on the same curve.
- Point doubling: An operation that combines a point with itself.
- Generator point: A predefined point used to generate other points on the curve.
- Private key: A secret numerical value owned by a user.
- Public key: A point on the elliptic curve derived from the private key.

These concepts form the foundation of ECC-based cryptographic systems.

How Does ECC Work?

ECC uses mathematical operations on elliptic curve points to establish a relationship between a private key and a public key.

The general process is as follows:

1. A suitable elliptic curve and generator point are selected.
2. A user generates a private key, which must remain secret.
3. The public key is generated by multiplying the generator point by the private key using elliptic curve point multiplication.
4. The public key can be shared with other users, while the private key is kept confidential.

The public key can be verified or used in cryptographic operations without revealing the private key.

Elliptic Curve Discrete Logarithm Problem

The security of ECC relies largely on the difficulty of the Elliptic Curve Discrete Logarithm Problem (ECDLP).

Suppose a public key is generated by multiplying a known generator point by a private key. It is computationally difficult to determine the private key from the generator point and the resulting public key when suitable curve parameters are used.

This one-way relationship makes ECC useful for secure communication and blockchain applications.

Importance of ECC in Blockchain

ECC plays an important role in blockchain systems because it supports:

- Generation of public and private key pairs.
- Digital signatures for blockchain transactions.
- Verification of transaction authenticity.
- Proof that a transaction was authorized by the holder of the corresponding private key.

For example, when a user initiates a blockchain transaction, the user signs the transaction using a private key. Other participants can verify the signature using the corresponding public key.

Advantages of ECC

- Strong security with relatively small key sizes.
- Efficient storage of cryptographic keys.
- Reduced communication overhead.
- Suitability for digital signatures and key exchange.
- Practical use in resource-constrained systems.

Summary

Elliptic Curve Cryptography is an important public-key cryptographic technique used in blockchain technology. It relies on elliptic curve point operations and the computational difficulty of the elliptic curve discrete logarithm problem. ECC enables secure key generation and supports digital signatures used to authenticate blockchain transactions.

Campus : Prestige Vihar, Scheme No. 74-C, Sector D, Vijay Nagar, Indore -452010 (M.P.) India
Phone : 0731-4013333, Fax : 0731-4013329 E-mail : info@piemr.edu.in, Web. www.piemr.edu.in